KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2020
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 1.5 hours FULL MARKS: 120

N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) "Pattern matching by regular expression is greedy" – Explain the term taking example to match the word 'There' from a string. (10)
   b) Given two strings $S1$ and $S2$ and three operations ADD, DELETE and EDIT. How many minimum operations are needed to convert $S1$ to $S2$? Derive the algorithm to answer the question. Apply the algorithm for $S1$ = "*abcdef*" and $S2$ = "*azced*". If each of the operation costs 1 unit, find the total cost to convert $S1$ to $S2$. (20)

2. a) "Minimizing perplexity is same as maximizing the probability" – Explain the term. (12)
   b) Consider the following corpus: (12)

   $\langle S\rangle abc\langle /S\rangle$
   $\langle S\rangle bac\langle /S\rangle$
   $\langle S\rangle abc\langle /S\rangle$

   Calculate the bigram probabilities $P(a|\langle S\rangle)$, $P(c|b)$ and $P(b|a)$.
   c) How do you handle the stop words for text classification? (06)

3. a) Given the following text classification for C_1 and C_2 (14)

   $PQR \rightarrow C_1$
   $PPS \rightarrow C_1$
   $PT \rightarrow C_1$
   $UVP \rightarrow C_2$

   Compute the most likely class for the text "*PPPUV*".
   b) Consider the following grammar in CNF (16)

   $S \rightarrow AB \mid BC$
   $A \rightarrow BA \mid a$
   $B \rightarrow CC \mid b$
   $C \rightarrow AB \mid a$

   Is '*baaba*' in $L(G)$? Explain your answer using CYK algorithm.

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) "POS tagging is a disambiguation task" – Justify the statement with example. (10)
   b) Consider the sentence: (08)
   "*It/PRP would/MD be/VB better/JJR to/TO book/?a/DT ticket/NN via/IN internet/NNP*".
   The word "*book*" is often used as *VB* or *NN*. Given the probabilities below, find the right POS tag for the word "race".

   $P(VB|TO) = 0.67$, $P(NN|TO) = 0.02$, $P(book|VB) = 0.50$,
   $P(book|NN) = 0.50$, $P(DT|VB) = 0.02$, $P(DT|NN) = 0.06$.

   c) What is prosody? Discuss about the three aspects of prosody. Discuss the effect of zero situation. (12)

5. a) Given a sequence of ice-cream observations (1 1 3) and an HMM $\lambda = (A, B)$ in the following figure, find the best hidden weather sequence $Q(like\ H\ H\ H)$. (12)

A		B	
$Start \rightarrow HOT$	0.25		
$Start \rightarrow COLD$	0.75	$P(1 \mid HOT) = 0.2$	$P(1 \mid COLD) = 0.5$
$HOT \rightarrow HOT$	0.60	$P(2 \mid HOT) = 0.35$	$P(2 \mid COLD) = 0.4$
$HOT \rightarrow COLD$	0.40	$P(3 \mid HOT) = 0.45$	$P(3 \mid COLD) = 0.1$
$COLD \rightarrow COLD$	0.55		
$COLD \rightarrow HOT$	0.45		

b) Define the term odds for logistic regression. Show that the observation should be labeled true if $\sum_{i=0}^{N} w_i f_i > 0$. (12)

c) What are the components of Markov chain? (06)

6. a) Discuss about the steps of text to waveform mapping in speech synthesis using Hourglass metaphor. (10)

b) What is Homograph disambiguation? What are the problems of CMU? How does UNISYN overcome the problems of CMU? (10)

c) What are the use of Question Answering System (QAS)? Discuss about the main phases of QAS architecture. (10)